The Dark Web, Deep Web and Darknets

Mateus Grandolfi Albuquerque

Internet Protocols - Spring 2024-25

Abstract

This paper breaks down two of the three main layers of the Web—the *Deep Web* and *Dark Web*—and shows how they work under the hood. On the open, searchable part of the internet, sites use the usual HTTP/HTTPS protocols and rely on DNS to map names to IP addresses. The *Deep Web* is much larger. It holds data that sits behind log-ins or paywalls, so search engines cannot reach it, but it still uses the same basic web technology and does not hide who you are.

The *Dark Web*, however, is different. It runs on special overlay networks called *darknets*—mainly Tor, I2P, and Freenet. These systems wrap every message in several layers of encryption and send it through a chain of volunteer nodes. That way, neither the user nor the server can see each other's real IP address.

This document explains how each *darknet* functions. It also lists what you can find in each layer: everyday databases and private intranets in the *Deep Web*, but also dark-market shops, whistle-blower drop boxes, and censored news mirrors in the *Dark Web*.

Finally, I look at the broader impact. *Darknets* help criminals sell drugs and stolen data, yet they also protect activists, journalists, and regular users from surveillance and censorship. Because of this double edge, any policy response has to fight abuse without removing the privacy benefits these networks provide.

Index

1 Introduction **1**

2 Web Layers: Deep Web vs. Dark Web **1**

3 Standard web infrastructure (Surface & Deep Web) **2**

4 Darknets: Architecture and Maintenance **3**

4.1 The Tor Network (Onion Routing) 3

4.2 I2P (Invisible Internet Project) and Garlic Routing 5

4.3 Freenet (Distributed Anonymous Storage) 7

4.4 Summary of darknet designs 8

5 Anonymity and Obfuscation Mechanisms in Darknets **9**

6 Content and services **9**

6.1 Deep Web Content 10

6.2 Dark Web Content and Hidden Services 11

7 Impact of Darknets on the Internet and Society **13**

7.1 Data Security and Privacy Implications 14

7.2 Criminal Activity Enabled by Darknets 14

7.3 Censorship Resistance and Freedom of Information 15

7.4 Whistleblowing and Political Dissent 15

8 Conclusion **16**

9 References **16**

1 Introduction

The World Wide Web is commonly envisioned in layers – from the easily accessible sites indexed by search engines to hidden networks requiring special software. These layers are often illustrated by the iceberg metaphor, where the visible tip represents the indexed Surface Web, beneath which lies the vast Deep Web, and further below, the secretive Dark Web.

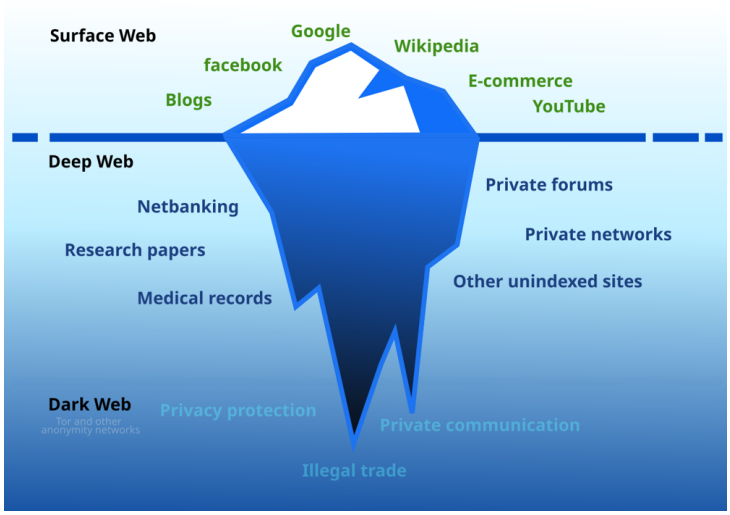

Figure 1: Conceptual "iceberg" of the web

The Deep Web consists of all the content that is not indexed by search engines, such as databases and exclusive websites. The Dark Web is a subsection of the Deep Web, intentionally hidden and accessible only with anonymity networks such as Tor, I2P, or Freenet. This paper discusses the distinction between the Deep and Dark Web and explores the technologies behind each level—from typical HTTP/HTTPS protocols to onion routing (Tor), garlic routing (I2P), and distributed storage of Freenet. It describes the architecture of darknets, node discovery, encryption, routing, and access to hidden services. Anonymity and obfuscation methods are analyzed, as well as content types seen in layers, ranging from academic sources to illicit marketplaces. The paper ends with an assessment of how darknets affect society and cybersecurity, considering problems like cybercrime, resistance to censorship, whistleblowing, and data security, according to industry and academic research.

2 Web Layers: Deep Web vs. Dark Web

The Deep Web includes all online content not indexed by search engines, making it invisible to conventional search. It is not necessarily sinister or clandestine; rather, it consists primarily of typical services such as secure mail, internet banking, academic databases, internal network websites, and cloud storage software. Access typically requires URLs or login credentials, but no special software. Though it operates over standard internet protocols (HTTP/HTTPS), it often uses encryption and authentication to secure information and restrict access to pubilc. Estimates by research put the Deep Web at about 90% of all web content.

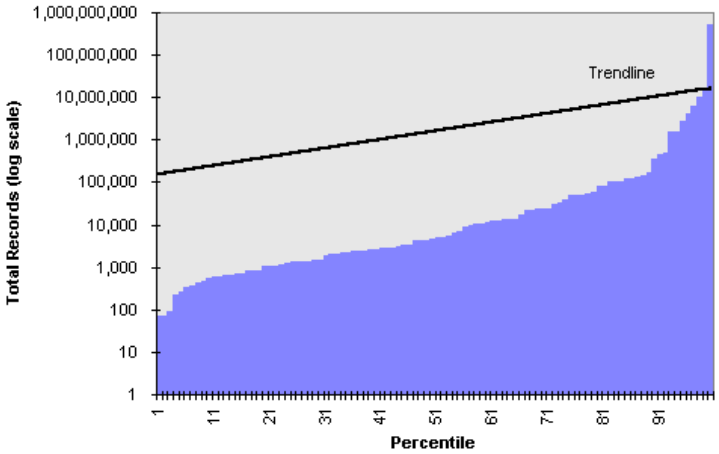

Figure 2: Inferred Distribution of Deep Web Sites, Total Record Size

The Dark Web, in comparison, is a deliberately hidden part of the Deep Web that requires specialized software, such as Tor, I2P, or Freenet, to access. Unlike typical Deep Web content, Dark Web sites operate on overlay networks with custom routing and addressing to preserve anonymity. Tor, the most well-known darknet, hosts .onion hidden services accessible only via the Tor browser. There's I2P and Freenet, among others, with .i2p sites known as eepsites, and Freenet as a peer-to-peer network for anonymous file sharing. Dark Web sites protect server and user identities through layered routing and encryption. An illustration of this is that the address of a Tor hidden service constitutes a hash derived from its public key, with all communication being encrypted and transmitted through various nodes to conceal IP addresses. Although the Dark Web is frequently linked to illegal activities because of its anonymity, it also serves legitimate functions such as facilitating privacy-sensitive communication and enabling resistance against censorship. Accessing them takes intent; users have to install darknet-specific software and know where to look, since content isn't searchable with conventional search engines.

3 Standard web infrastructure (Surface & Deep Web)

Before examining darknet architectures, it's important to understand how the surface and deep web work, as both rely on core internet protocols that darknets build from:

- TCP/IP and Routing: All web traffic uses the Internet Protocol (IP), routed through networks via BGP and autonomous systems. Packets are forwarded hop-by-hop toward the destination IP. This system is transparent: source and destination IPs are visible to any node along the path, even if the payload is encrypted.

- DNS and Addressing: Surface web domains are resolved through the public DNS. Deep web content often also uses DNS but may be unlisted or require credentials (e.g., *intranet.company.com*). Dark web addresses, such as .onion, bypass public DNS entirely and are resolved internally within darknets like Tor.

- HTTP/HTTPS: Both surface and deep web use HTTP or its encrypted form, HTTPS. Deep web content (e.g., behind logins or paywalls) may use HTTPS but is inaccessible to search engines. HTTPS encrypts data but still reveals metadata like the server's IP. Darknets go further by hiding both endpoints.

- Indexing and Crawling: The surface web is defined by being crawlable. Web crawlers[1] index publicly linked pages, while the deep web includes content behind forms, logins, or paywalls—anything not discoverable by bots. This isn't a protocol distinction but one of architecture and access control.

4 Darknets: Architecture and Maintenance

Darknets are overlay networks designed specifically for anonymity and privacy. They operate on top of the Internet but employ custom protocols for routing and encryption. I will examine three major darknets – Tor, I2P, and Freenet – covering how each constructs its network, discovers and routes to nodes, and maintains security.

4.1 The Tor Network (Onion Routing)

- Node Discovery and Directory: Tor keeps track of its thousands of volunteer relays through many trusted directory authorities. Every hour, each relay sends a signed descriptor (IP, public keys, bandwidth, and so on) to these authorities, which compile the data into a single "consensus" list of active, trusted relays. When you start Tor, your client downloads this consensus to know which relays to use. Coordination is centralized in those few authorities, but the consensus itself is copied across the network for resilience.

- Circuit Construction: The Tor client builds a circuit through three relays: an entry (guard) node, a middle relay, and an exit node. The entry node sees the client's IP, the exit node connects to the final destination, and the middle relay links the two. Relays are selected from the consensus, weighted by bandwidth and stability. Crucially, no single node knows the entire path: the entry knows the user but not the destination, the exit knows the destination but not the user, and the middle relay knows neither. This separation preserves anonymity.

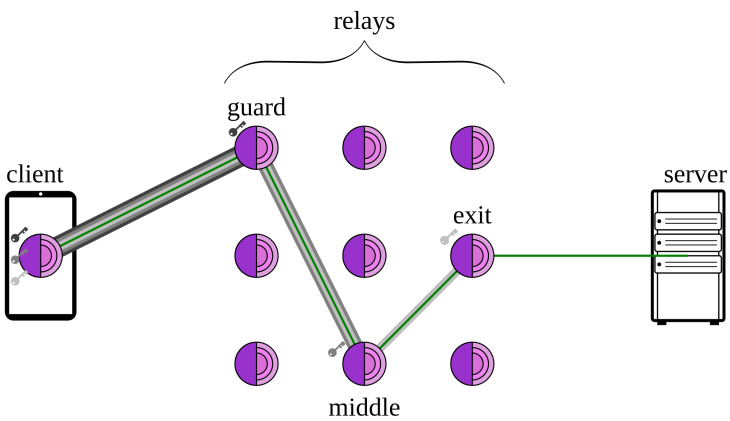

Figure 3: Tor Routing

- Onion Encryption: Tor wraps every message in three layers of encryption—exit, middle, entry—using a separate symmetric key for each relay. The client sends this "onion" to the entry relay, which peels its layer, learns only the next hop, and forwards the rest. Each relay repeats the process until the exit removes the last layer, revealing the plaintext and destination. The exit fetches the web page (or, for

[1] https://en.wikipedia.org/wiki/Web_crawler

hidden services, meets at a rendezvous relay) and routes the reply back through the same path, re-encrypting at each hop. Because no relay ever sees both your IP address and your data, and outside observers see only shuffled, encrypted packets, tracing Tor traffic is extremely difficult.

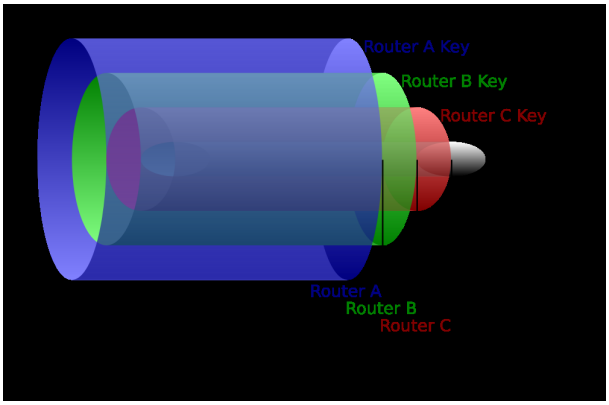

Figure 4: Onion routing encryption layers

- Hidden Services (Onion Services): Tor supports hidden services; servers that protect their IP addresses and are only accessible via the Tor network through a .onion address. This address is a hash of the service's public key. To host a hidden service, the server publishes descriptors to a distributed hash table (DHT)[2], listing introduction points (Tor relays). A client who wants to connect retrieves the descriptor, contacts an introduction point, and both client and service create circuits to a chosen rendezvous point—a separate relay where they connect anonymously. Throughout this process, onion routing is used: data is encrypted in layers, and no relay knows both identities.

- Circuit Rotation and Ephemeral Keys: Tor circuits have a short life span. A single circuit is typically used for at most 10 minutes or a set of connections; then a new circuit is built. This limits the amount of data a relay sees and prevents long-term correlation. Also, the Diffie-Hellman key exchanges[3] ensure perfect forward anonymity on each circuit. If a user somehow compromises one circuit's keys, it doesn't help to decrypt past or future circuits.

- Encryption and Authentication: Tor uses multi-layer encryption for strong privacy. Traffic is encrypted in layers, and TLS[4] is used between relays to secure link-level communication from outside observers. This ensures both data and relay-to-relay connections are protected. Tor operates at the TCP stream level, tunneling TCP connections through the chosen circuit. It does not natively support UDP, except for limited DNS resolution via exit nodes.

- Performance and Maintenance: The Tor network requires continual maintenance by its community – updating the directory consensus, adding or removing relays, and adjusting algorithms to balance load. As of recent years, Tor has over 6000–7000 relays and several million users globally. Tor developers also implement various defenses against known attacks and provide auxiliary services like bridges (unlisted entry relays to help avoid Tor censorship by ISPs or governments).

[2] https://en.wikipedia.org/wiki/Distributed_hash_table
[3] https://en.wikipedia.org/wiki/Diffie%E2%80%93Hellman_key_exchange
[4] https://www.cloudflare.com/learning/ssl/transport-layer-security-tls/

4.2 I2P (Invisible Internet Project) and Garlic Routing

- Pure Peer-to-Peer Topology: I2P uses a pure peer-to-peer model where every user acts as both router and client. Each node routes traffic and can host services called eepsites. Unlike Tor, which separates clients from relays and relies on central directories, I2P is fully decentralized with no central servers.

- Distributed Network Database (NetDb): Instead of Tor's central directories, I2P uses a distributed database called NetDb, based on a modified Kademlia DHT[5] run by special "floodfill" routers. Nodes publish their info to these peers and query the NetDb to find others or services. This decentralized system requires enough floodfill nodes to maintain the network, with new nodes starting from known peers to join and participate in routing.

- Unidirectional Tunnels: I2P uses one-way tunnels for routing, unlike Tor's bidirectional circuits. To enable two-way communication, I2P sets up four tunnels: outbound and inbound tunnels for both sender and receiver. Each tunnel consists of a series of routers (typically 3 by default). For example, if Alice wants to access Bob's eepsite, she sends data through her outbound tunnel to Bob's inbound tunnel, using Bob's gateway info from the network database (NetDb). Bob's response returns via his outbound and Alice's inbound tunnel. This unidirectional tunnel structure improves resistance to traffic analysis by separating send and receive paths and is a core distinction of garlic routing (used by I2P) compared to onion routing (used by Tor).

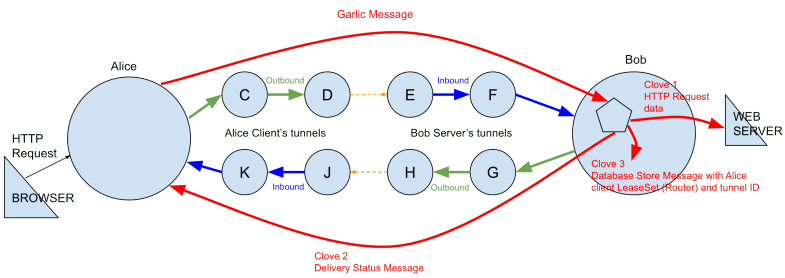

Figure 5: I2P's tunnel routing

- Tunnel Building: To create a tunnel, an I2P router selects a sequence of other routers and sends them tunnel build messages. Each router in the sequence is only aware of the next hop (like onion routing). Tunnels are built periodically and last for a short time (I2P tunnels typically expire after 10 minutes). The short lifespan and continuous rotation of tunnels help anonymity by limiting exposure. The capacity of tunnels (how many messages can flow) is also managed. If a tunnel is about to expire or is underutilized, new ones are built and old ones are torn down – this is part of I2P's self-maintenance.

[5] https://geti2p.net/en/docs/how/network-database

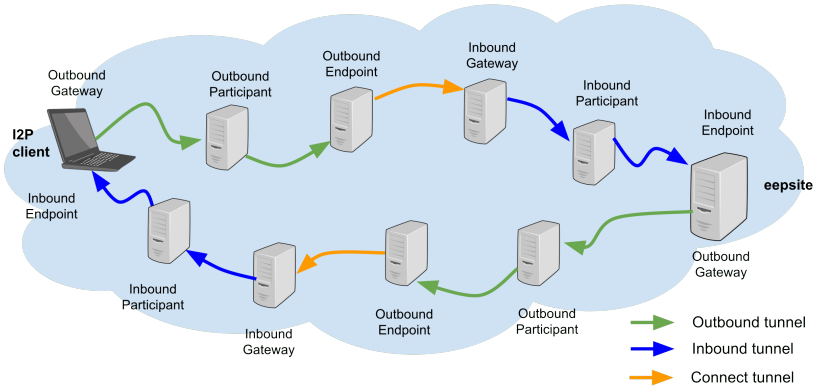

Figure 6: I2P's tunnel building

- Garlic Routing: Garlic routing, I2P's variant on onion routing, wraps several independent messages ("cloves") inside one encrypted packet ("bulb"). Each clove can carry its own delivery instructions (possibly different destinations or reply paths) so a single outbound tunnel can transport multiple requests together. This boosts throughput and hinders traffic analysis, since observers see only one ciphertext carrying many layered payloads. Although I2P rarely exploits every multi-recipient feature, it does merge control tokens and data in the same bulb, cutting overhead while preserving anonymity.

- Encryption and Anonymous Routing: I2P applies layered encryption to each hop in its one-way tunnels. When Alice injects a garlic packet into her outbound tunnel, every router removes a layer until the packet exits and reaches Bob's inbound gateway, which then forwards it along Bob's inbound tunnel. Throughout the trip, the payload is also end-to-end encrypted with Bob's public key, so only Bob can read it (intermediate nodes see routing info but not the data itself). Because outbound and inbound tunnels are unidirectional and unique, Alice's request and Bob's reply travel over different routes, forcing an attacker to correlate two separate paths.

- Addressing and "Destinations": I2P skips DNS and assigns each service a cryptographic "Destination", essentially a public-key pair shown as a 52-character .b32.i2p string[6]. Local address books can map that string to a friendlier .i2p name. When Alice requests example.i2p, her router resolves it to the Destination, fetches its leaseSet from NetDb (a list of current inbound gateways), then garlic-encrypts the packet and pushes it through her outbound tunnel to one of those gateways.

- Community and Maintenance: I2P is open-source and grows with user participation: every node donates bandwidth, so more users mean more capacity. Traffic moves over TCP and a custom UDP protocol (Secure Semi-reliable UDP) which handles NAT traversal and boosts speed; its improved SSU2[7] variant debuted in 2022 for better security and performance. Because the network is fully peer-to-peer with no central authority, user counts are estimated from data that routers share with floodfill peers.

[6] https://geti2p.net/spec/b32encrypted
[7] https://geti2p.net/spec/ssu2

4.3 Freenet (Distributed Anonymous Storage)

- Data-Centric Model: Freenet works as a giant, encrypted cache. Uploaded files are split, encrypted, and scattered across peers; each chunk is referenced by a content key or keyword. To retrieve a file, a user sends that key into the network, which hunts down the pieces and reassembles them. Popular files replicate automatically, staying online even if the original publisher disappears. No single server ever hosts the full content.

- Routing Algorithm: Freenet's routing is based on a heuristic key-based routing[8]. Each Freenet node keeps a table linking nearby peers to the content keys they have served before. When a user requests a key, the node checks its own cache; if missing, it forwards the request to the neighbor whose key hash is "closest" to the target. The query keeps propagating (backtracking if it hits a dead end) until the data is found or a hop limit is hit. Replies follow the same path back, and every node along the way may cache the chunks, making future requests quicker. Throughout, no node sees both source and destination, maintaining anonymity.

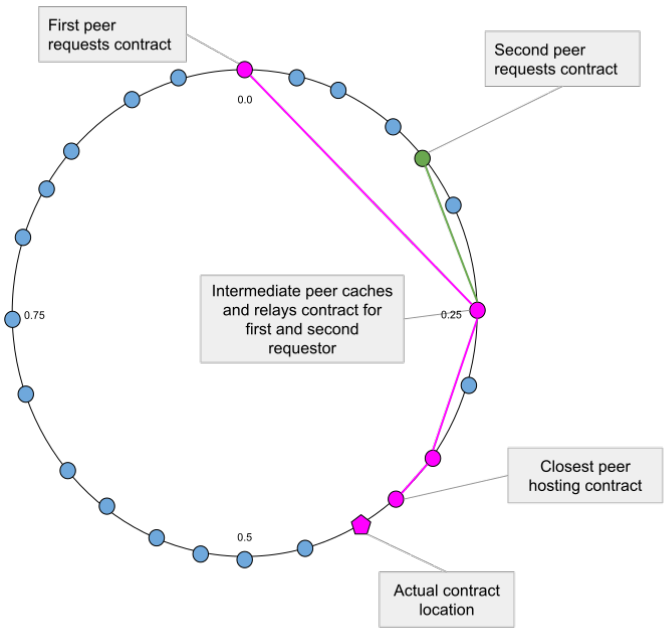

Figure 7: Freenet routing scheme

- Anonymity: Freenet protects both publishers and readers. When a user uploads content, it is broken into encrypted chunks, each tagged with a key, and distributed to nodes whose IDs are closest to those keys. After insertion, the data is separated from its originator and simply "lives" in the network. Readers retrieve files by key: their requests traverse several nodes, and each node cannot tell whether the request began with its predecessor or was merely forwarded, preserving requester anonymity. Nodes store only encrypted fragments, often without knowing what they hold or who published it, giving operators plausible deniability. Even if a node caches illicit data, the operator can truthfully claim it arrived automatically via the protocol.

[8] https://en.wikipedia.org/wiki/Heuristic_routing

- Darknet Mode (Friend-to-Friend): Freenet can run as a closed web of trust where a node connects only to peers it has manually added (keys exchanged offline). This makes the network extremely hard to map or infiltrate, offering strong anonymity and censorship resistance, though performance and search reach diminish if the friend group is small. Freenet also provides opennet, which auto-connects to random peers via public seed nodes, which is simpler but more vulnerable to monitoring and blocking. Users often start in opennet and gradually shift to friend links.

- Content Persistence and Eviction: In Freenet, every request caches encrypted chunks along its route, so popular files quickly replicate across the network, while not very accessed data is rid to free space. This means that once a file attracts even a modest audience, its fragments become dispersed across thousands of volunteer nodes, making it virtually impossible to erase and giving Freenet powerful, decentralized resistance to censorship.

- Data Encryption and Security: All Freenet traffic moves through encrypted links (first over a custom TCP protocol and, in newer versions, UDP as well) and the data itself is encrypted. Each file is stored under a Content Hash Key[9] that pairs a hash with its encryption key; without that key, intercepted chunks are unreadable. Nodes reveal only their "next-door" neighbors, so tracing a request requires hopping from peer to peer.

- Use Cases and Services: Freenet hosts its own suite of applications, including static or dynamic "Freesites", discussion boards, file-sharing hubs, and pseudonymous email. A Freesite is just a collection of files stored under a site key that users retrieve with that key, and publishers update the site by releasing new, versioned keys.

4.4 Summary of darknet designs

In summary, Tor, I2P, and Freenet represent three different types of darknets:

- Tor: Low-latency onion routing overlay for both anonymous browsing and hidden services. It uses a mix of centralized and decentralized elements. Mainly circuit-based routing, good for interactive traffic.

- I2P: Peer-to-peer garlic routing network focusing on internal services. Fully decentralized participant routers, unidirectional tunnels, and DHT for peer discovery. Low latency for hidden services.

- Freenet: Peer-to-peer distributed storage system for censorship-resistant publishing. Prioritizes anonymity and persistence over latency. Uses key-based routing and caching, with an option for friend-to-friend connections. Suitable for forums or file sharing, but not real-time web access.

All three use encryption and clever routing to hide identities. Next, I will examine what other obfuscation techniques are used across darknets.

[9] https://github.com/hyphanet/wiki/wiki/Content-Hash-Key

5 Anonymity and Obfuscation Mechanisms in Darknets

True anonymity online goes beyond basic encryption. It must hide identities, traffic patterns, and any link between actions and people. Darknets achieve this with layered encryption, relay chains, decentralized networks, and protocol-level obfuscation. Some of these mechanisms include onion encryption, proxy relays and routing hops, and peer-to-peer decentralization, etc. The sections that follow explain some other mechanisms and how one of them works:

- Encryption of Metadata and Padding[10]: Darknets encrypt not just the content of a message but almost all of the surrounding metadata. In Tor, every relay sees only the information it needs (the next hop) while the rest of the header stays encrypted. I2P's SSU transport applies the same principle to its router handshakes, and Freenet hides true content keys behind hashes so intermediaries never learn them. Many networks also add dummy traffic to distorsion flow patterns: I2P can make tunnels with extra packets, and Tor increasingly relies on mechanisms like obfs4[11] or meek[12] that disguise Tor traffic as ordinary web or CDN connections.

- Anonymity vs. Performance: More privacy usually means slower speeds. I2P lets you choose quick, low-anon tunnels or slower, safer ones; Freenet offers smooth "opennet" access or a friends-only darknet; Tor can add bridges for extra cover. All three mix traffic to confuse eavesdroppers[13], yet timing attacks have still broken Tor in the past.

- No Trust in Exit (Encryption for Application Data): Tor's exit relay can read or tamper with anything you send in plain HTTP, so you should still use end-to-end encryption (HTTPS, TLS mail, encrypted chat) on top of the darknet. Many .onion sites already serve pages over HTTPS, giving Tor anonymity plus payload encryption; I2P traffic is garlic-encrypted inside the network, but apps often add their own layer too. Stacking network-level and application-level encryption is the safest way to keep both identity and data secure.

In summary, darknets stay private by bouncing data through several nodes, encrypting every hop, scattering knowledge so no one sees the full path, and adding random noise. These tricks hide who's talking to whom, when, and how much. It's far harder to trace than the regular web, where one ISP or server can log everything.

6 Content and services

The types of content one finds on the Deep Web and Dark Web differ significantly due to the nature and intended use of each layer. I will outline what content resides in each. Understanding this content breakdown also highlights how and why different layers are used.

[10] https://en.wikipedia.org/wiki/Padding_(cryptography)
[11] https://support.torproject.org/glossary/obfs4/
[12] https://gitlab.torproject.org/legacy/trac/-/wikis/doc/meek
[13] https://en.wikipedia.org/wiki/Eavesdropping#Network_attacks

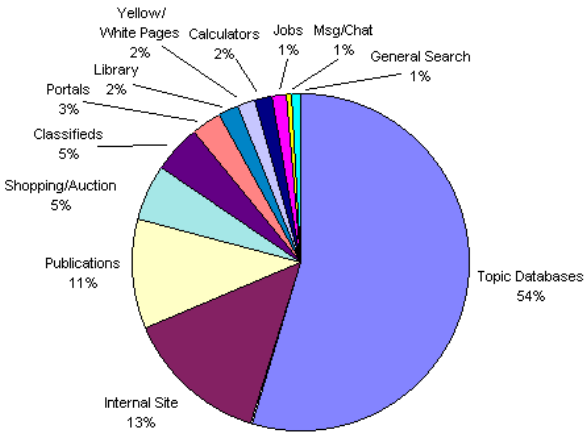

Figure 8: Distribution of Deep Web Sites by Content Type

6.1 Deep Web Content

The deep web contains a huge range of content. Some include:

- Private Databases and Web Applications: Many websites have front-end search forms that query databases (e.g., airline flight databases, library catalogs, patent databases). For instance, a library's search result for "quantum physics" might be a deep web page because you must query it specifically.
- Academic Journals and Subscription Content: Paywalled content, such as scientific journal articles which require subscription or institutional access. Search engines might index the abstract or title, but the full PDF lies in the deep web, accessible only after login/payment.
- Email, Cloud Storage, and Account Data: Personal Gmail inbox or Google Drive files are deep web – you must login to view them; they are not publicly listed. Similarly, your private social media messages content is deep web.
- Online Banking and Financial Records: When you log into your bank account portal and view statements, those pages are generated for you and not accessible to others or search engines.
- Internal Corporate or Government Sites: Intranets, portals... that require VPN or special access. For example, a company wiki or a government personnel site; these are deep web by virtue of being closed to the public internet.
- Unlisted Web Pages: Any page that is not linked from the surface web and not meant to be found. For example, a raw upload or backup file on a web server that isn't linked on the site. If it's not disallowed by robots.txt[14] and not linked, it might remain unindexed.
- "Dark Web" content accessible via Gateways: Interestingly, some dark web content can be partially accessed via special gateways on the surface web (like Tor2Web proxies which allow someone to visit an .onion through a normal browser via a relay). In those cases the content is still considered dark web, but the specific gateway page on the clearnet could be seen as a deep web proxy.

[14] https://en.wikipedia.org/wiki/Robots.txt

6.2 Dark Web Content and Hidden Services

The dark web is notorious for certain kinds of content, and while not all its content is illicit or unethical, a substantial portion has leaned toward uses that prefer to avoid identification. Here are major categories:

- Illicit Marketplaces: Dark web marketplaces (pioneered by Silk Road[15] (2011-2013)) use Tor hidden services and cryptocurrencies to let people trade drugs, stolen data, fake documents, hacking tools, and even weapons. Police have taken down some of these websites, yet replacements keep emerging. A 2016 study found that 57% of surveyed dark web sites offered illicit goods, mainly drugs, followed by financial fraud and hacking services.

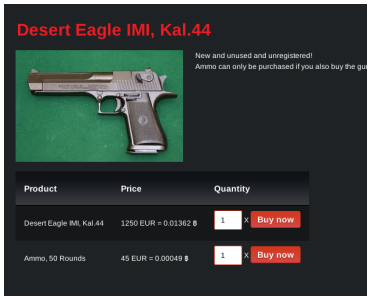

Figure 9: Desert Eagle at sale on the EuroGuns hidden service

Figure 10: Cannabis buds offered on the German DeDope marketplace

- Cybercrime Forums and Services: Dark web forums on networks like Tor and I2P serve as meeting grounds for cybercriminals, where they trade malware, ransomware partnerships, stolen credit cards... Stolen credentials have become a cheap commodity: in 2022 an estimated 15 billion usernames and passwords were circulating, and researchers recently counted more than 22 000 listings of personal data: credit cards, IDs, online accounts... adding up to hundreds of thousands of sales.

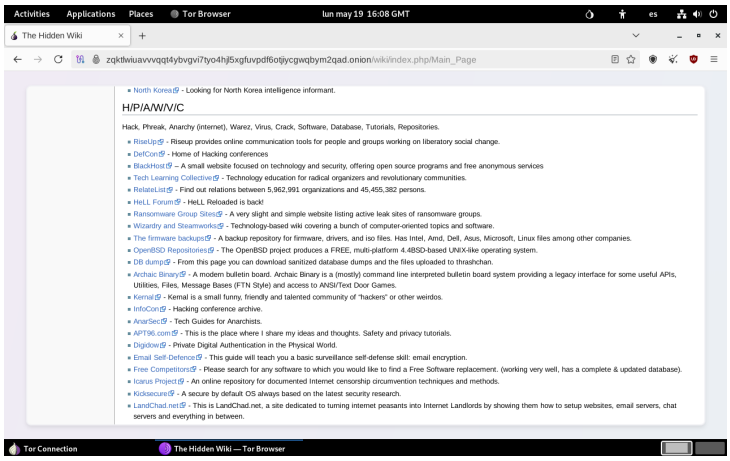

Figure 11: HiddenWiki links related to cybercrime

[15] https://en.wikipedia.org/wiki/Silk_Road_(marketplace)

- Illegal Pornography and Exploitation Content: Some dark web sites hide content that's outright illegal; like child-exploitation material. These hidden services are fewer than drug markets but cast a long shadow over the darknet's reputation. Law-enforcement teams actively hunt them down; the FBI's 2015 takedown of the Playpen site[16] is a well-known example. The anonymity Tor (among many) offers lets offenders share this material with far less risk of immediate discovery.

- Extremist and Terrorism-related Sites: Extremist groups sometimes host propaganda and forums on Tor to dodge takedowns (jihadist sites and lone-wolf manifestos have appeared this way) but they also spread material via encrypted apps and disguised surface in web pages, making the dark web just one of several channels.

- Anonymity-focused Services: Some dark web sites exist to defend privacy, not commit crimes. SecureDrop nodes let whistle-blowers hand documents to journalists anonymously, and people in repressive countries reach forums that censors can't block. Major services mirror themselves for the same reason: Facebook launched an .onion site in 2014, and outlets like The New York Times, BBC, Deutsche Welle, and ProPublica publish via Tor so readers can still access news and send tips.

Figure 12: SecureDrop's website

- Political Discussion and Whistleblower Platforms: Tor hosts anonymous drop sites where whistle-blowers can post documents without exposing their identity or location. Activists and dissidents also use Tor or I2P forums to plan and share information when the regular web is watched or blocked.

[16] https://en.wikipedia.org/wiki/Playpen_(website)

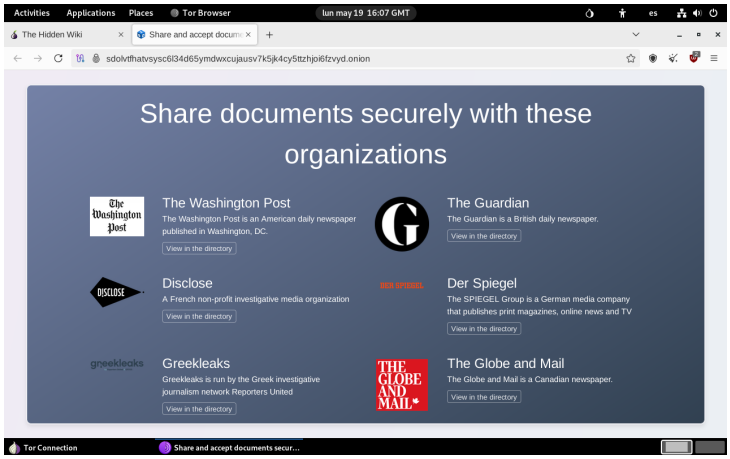

Figure 13: SecureDrop's document-sharing portal for whistleblowing

- Financial Services and Fraud: Some dark web services facilitate money laundering or mixing of cryptocurrencies to break exposure. Additionally, there are scam sites, phishing pages, or fake services aiming to steal cryptocurrency from unsuspecting dark web users (crime targeting criminals, ironically).

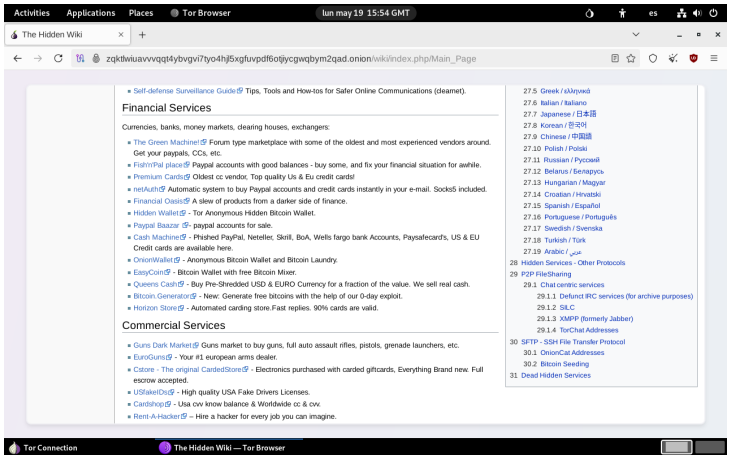

Figure 14: HiddenWiki's main financial services links of interest

- Pirated Content and File Sharing: Tor and I2P host hidden sites for pirated books, music, and movies, but heavy file sharing usually shifts to BitTorrent (either on the open internet or over I2P) because Tor's bandwidth is limited. Freenet, with its distributed storage, often carries banned or controversial literature instead.

- Emerging Services: New tools are tying decentralization to darknets. Namecoin offers .bit domains[17] that Tor can look up without relying on regular DNS, while peer-to-peer messengers such as Ricochet and Cwtch run over hidden services so no metadata leaks out.

7 Impact of Darknets on the Internet and Society

Darknets reshape the internet: they push stronger security habits, open doors for new kinds of online crime, let users slip past censorship, and give whistle-blowers and activists safe channels to speak. In the following sections, I will evaluate some of these aspects.

[17] https://peername.com/dot-bit/

7.1 Data Security and Privacy Implications

Darknets affect data security in complex ways. On the positive side, they enhance privacy; journalists, activists, and others use Tor or encrypted services to avoid surveillance, and the privacy techniques developed for darknets have inspired mainstream improvements like widespread HTTPS and encrypted DNS[18]. Organizations even monitor dark-web forums to detect breaches early.

But darknets also create marketplaces for stolen data, hacking tools, and malware, worsening the cybersecurity landscape. Stolen information from breaches quickly appears on dark web markets, making identity theft and fraud easier. Cybercrime tutorials and "services" offered online lower the entry barrier for attackers, further increasing threats.

Law enforcement has had to adapt, using digital forensics, malware, and blockchain analysis to catch dark web criminals, triggering a continual cat-and-mouse game of evolving tactics on both sides.

7.2 Criminal Activity Enabled by Darknets

Darknets have a notoriety for enabling criminal activity that would be far riskier or impossible on the public internet. The anonymity and global reach of darknets expand the scale and scope of certain crimes:

- Online Drug Trade: Dark web markets like Silk Road transformed the drug trade by moving it online, offering anonymity, easy transactions, and customer reviews. The scale became significant enough for major police operations, such as Operation Onymous[19] in 2014, shutting down many Tor-based markets. Despite repeated crackdowns, new markets keep emerging.

- Fraud and Identity Theft: Dark web markets have made identity theft and fraud global and easy. Before, stolen data traded in small circles; now anyone using Tor can cheaply buy credit cards or fake IDs online, contributing to massive fraud worldwide.

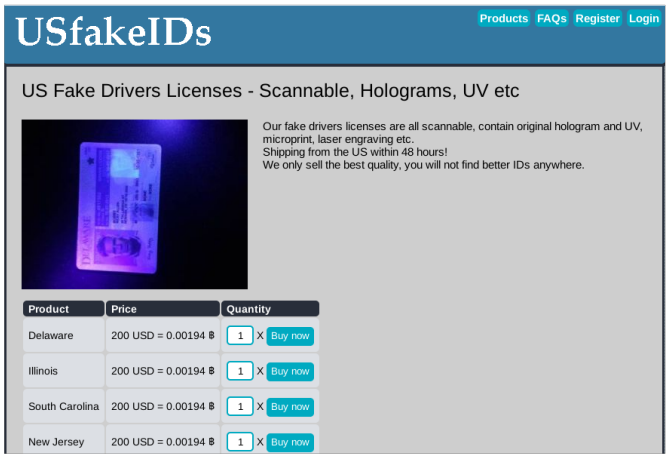

Figure 15: Fake US-based driving licenses seller

- Malware and Cyberattacks: The dark web helps with cyber-attacks by selling malware, zero-day exploits, and even remote access to hacked machines. Botnet[20]

[18] https://en.wikipedia.org/wiki/DNS_over_HTTPS
[19] https://www.europol.europa.eu/operations-services-and-innovation/operations/operation-onymous
[20] https://en.wikipedia.org/wiki/Botnet

and ransomware gangs hide their command-and-control servers[21] behind Tor, making them hard to shut down. Ransomware-as-a-Service kits are rented for a cut of the profits. Attacks on hospitals, businesses, and governments have happened in recent years.

- Financial Crime & Money Laundering: Cryptocurrency powers nearly all dark web commerce. Bitcoin's earliest boom came from markets like Silk Road. Criminals now launder funds by swapping them for privacy tokens, services frequently advertised on Tor.

Because dark-web crime jumps borders, takedowns demand multi-nation teamwork: AlphaBay's 2017 bust[22], for instance, pulled in authorities from Thailand, the U.S., Canada, and France. Jurisdictional gaps slow investigations, forcing agencies to join forces.

7.3 Censorship Resistance and Freedom of Information

Darknets are vital tools for breaking through state censorship. In countries like China, Iran, or Russia, Tor lets citizens slip past firewalls, a fact reflected in user spikes during protests. Bridge nodes, whose IPs are hidden, keep Tor reachable even when known relays are blocked. Media outlets mirror themselves on .onion sites for the same reason: the BBC, New York Times, and others use Tor to reach readers in censored regions. Activists, whistle-blowers, LGBTQ+ groups, and citizen journalists use hidden forums and SecureDrop-style services to communicate and share documents safely. By giving people a channel for uncensored news and private conversation, darknets act as a modern samizdat[23] and preserve the internet's freedom ideal. Their success, however, has made them targets; China deviates Tor traffic and Russia has tried to block it, yet obfuscation tools keep them alive. In short, darknets provide an essential escape route for free expression wherever the surface web is filtered or surveilled.

7.4 Whistleblowing and Political Dissent

Darknets give whistle-blowers and dissidents a safer way to speak. Snowden defended Tor, and after his leaks, many newsrooms installed SecureDrop so sources can pass files without revealing who they are. Activists in places like Belarus (2020)[24] and Iran (2009)[25] have relied on Tor or I2P to organize when the open internet was watched or cut off, while hidden-service blogs let authors publish banned opinions that censors struggle to erase.

This anonymity encourages leaks that expose corruption and human-rights abuses, and hacktivists sometimes host drop sites that feed investigations and reforms. This system isn't perfect (operational missteps or forged leaks can still cause harm) but it shifts power: even tightly controlled regimes can't be sure they've silenced every voice.

[21] https://www.techtarget.com/whatis/definition/command-and-control-server-CC-server
[22] https://www.fbi.gov/news/stories/alphabay-takedown
[23] https://www.britannica.com/technology/samizdat
[24] https://en.wikipedia.org/wiki/Censorship_in_Belarus#2020s
[25] https://en.wikipedia.org/wiki/2009_Iranian_presidential_election_protests#Censorship

8 Conclusion

Looking across the internet's three layers (Surface, Deep, and Dark) it's clear each exists for a reason, yet each brings its own tensions. The Surface Web gives us convenience and visibility but little privacy. The Deep Web keeps all of our private data behind log-ins, reminding us that most information was never meant for search engines to begin with. The Dark Web, accessible only through networks like Tor, I2P, and Freenet, pushes the idea of privacy to its limits: it enables black-market trade and ransomware operations, yet it also keeps news flowing into censored countries and lets whistle-blowers reach journalists without fear.

Darknets illustrate the constant give and take between anonymity and accountability. They create threats like online drug sales and massive data breaches, yet they also keep the light on in places where free speech would otherwise be inexistent. As more of life moves online, the demand for privacy will only grow. Understanding how these networks work is no longer optional for policymakers, security professionals, or anyone who cares about the future of an open, yet secure, internet.

The dark web will likely remain a part of the internet's ecosystem, reflecting humanity's division: it keeps both our darkest activities and our most critical needs for freedom. We must continue to peel back the layers (much like the onion itself) to show how these networks function and how we can coexist with them in a way that maximizes safety, security, and liberty in the digital age.

9 References

References

Beshiri, A. S., & Susuri, A. (2019). *Dark web and its impact in online anonymity and privacy.* https://www.scirp.org/journal/paperinformation?paperid=91242

BrightPlanet. (2001). *The deep web: Surfacing hidden value.* https://quod.lib.umich.edu/j/jep/3336451.0007.104?view=text;rgn=main

Clarke, I., Sandberg, O., Wiley, B., & Hong, T. W. (2000). *Freenet: A distributed anonymous information storage and retrieval system.* https://www.freehaven.net/anonbib/cache/freenet.html

Clarke, I., Sandberg, O., Wiley, B., & Hong, T. W. (2001). *Freenet: A distributed anonymous information storage and retrieval system (pdf).* https://www.cs.cornell.edu/people/egs/615/freenet.pdf

CrowdStrike. (2025). *The dark web explained.* https://www.crowdstrike.com/en-us/cybersecurity-101/threat-intelligence/dark-web-vs-deep-web/

Cyble Research & Intelligence Labs. (2025). *Top 7 dark web marketplaces of 2025.* https://cyble.com/knowledge-hub/top-dark-web-marketplaces-of-2024/

Encora. (2022). *Detect and prevent enterprise risk from the dark web.* https://www.encora.com/insights/detect-and-prevent-enterprise-risk-from-dark-web

Freenet Project. (2025). *Freenet help page.* https://staging.freenetproject.org/pages/help.html

Greening, J. (2025). $17.3MinStolenPersonalDataSoldontheDarkWeb$. https://www.gasa.org/post/alarming-17-3m-trade-in-stolen-personal-data-on-the-dark-web

Jayasekara, D. (2016). *Deep dive into tor: The onion router*. https://blog.insiderattack.net/deep-dive-into-tor-the-onion-router-6de4c25beba7

Max Planck Institute. (2001). *Deep web white paper*. https://resources.mpi-inf.mpg.de/d5/teaching/ws01_02/proseminarliteratur/deepwebwhitepaper.pdf

Muncaster, P. (2024). *Don't become a statistic: Keep your personal data off the dark web*. https://www.welivesecurity.com/en/cybercrime/dont-become-statistic-defending-data-dark-web/

Navigating-the-Darkweb Project. (2023). *Chapter 3: I2p*. https://navigating-the-darkweb.readthedocs.io/en/latest/chapter3_i2p.html

Pangarkar, T. (2024). *Dark web intelligence market to hit usd 2.9 billion by 2032*. https://scoop.market.us/dark-web-intelligence-market-news/

Raman, R., Nair, V. K., Nedungadi, P., Ray, I., & Achuthan, K. (2023). *Darkweb research: Past, present, and future trends*. https://pmc.ncbi.nlm.nih.gov/articles/PMC10695971/

Sophos — Naked Security. (2019). *New bbc 'dark web' tor mirror site aims to beat censorship*. https://news.sophos.com/en-us/2019/10/28/new-bbc-dark-web-tor-mirror-site-aims-to-beat-censorship/

The Guardian. (2025). *Securedrop*. https://www.theguardian.com/securedrop

Tran, B. (2025). *Data breach statistics: How many records are leaked per year?* https://patentpc.com/blog/dark-web-monitoring-how-often-is-leaked-data-found

Wikipedia Contributors. (2025a). *Garlic routing*. https://en.wikipedia.org/wiki/Garlic_routing

Wikipedia Contributors. (2025b). *Onion routing*. https://en.wikipedia.org/wiki/Onion_routing

Wikipedia Contributors. (2025c). *Tor (network)*. https://en.wikipedia.org/wiki/Tor_(network)

Williams, J. (2016). *Research reveals 57% of dark web is illicit material*. https://www.tweaktown.com/news/50165/research-reveals-57-dark-web-illicit-material/index.html

Zaballos, E. (2025). *The dark web: Where stolen data becomes currency*. https://www.scworld.com/perspective/the-dark-web-where-stolen-data-becomes-currency